

Activity-4 Hashing problem.

Primary hash function is $h(x) = x \text{ mod } 10$

Input order = 4371, 1323, 6173, 4199, 4344, 9679, 1989.

initial hash value

Key $x \text{ mod } 10$

4371 - 1

1323 3

6173 3

4199 9

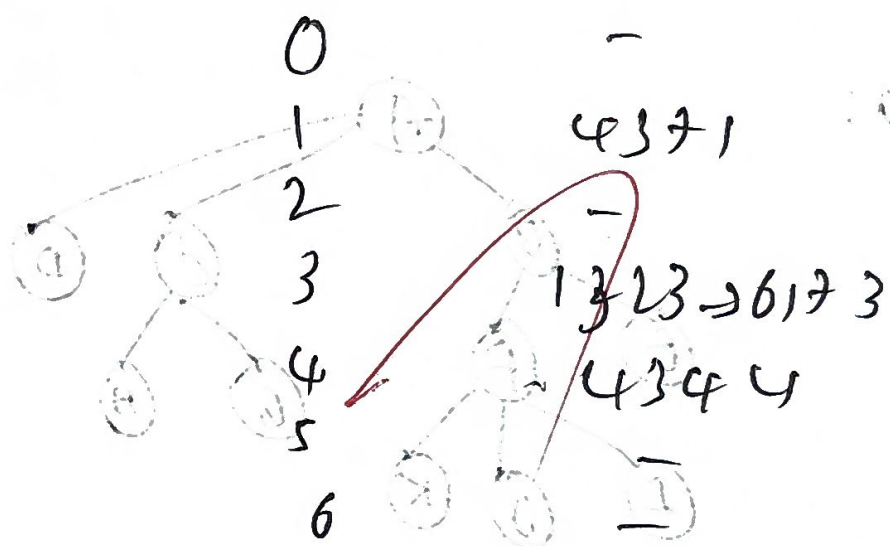
4344 4

9679 9

1989 9

a. Separate chaining

Index Chain



b. Open Addressing Linear probing.

Formula: $h_i(x) = (h(x) + i) \text{ mod } 10$ Final table:

4371 → index 1

1323 → index 3

6173 → 3 occurred → 4,

4199 → 9

4344 → 4 occurred → 5

9679 → 9 occurred → 0

1989 → 9 occurred → 10

Index	Value
0	9679
1	4371
2	1989
3	1323
4	6173
5	4344
6	-
7	-
8	-
9	-

C. Quadratic probing

der
 $h(x) = (h(x) + i^2) \text{ modulo } 10$

4321 $\rightarrow 1$

1323 $\rightarrow 3$

6173 $\rightarrow 3 \rightarrow 0 \text{ cells} + 1^2 = 4$

4199 $\rightarrow 9$

4344 $\rightarrow 4 \text{ cells} \rightarrow 4 + 1^2 = 5$

9679 $\rightarrow 9 \text{ cells} \rightarrow 9 + 1^2 = 0$

1989 $\rightarrow 9 \text{ cells}$

$\because 9 + 1 = 0, 0 \text{ cells}$

$9 + 4 = 3$

$9 + 9 = 0, \text{ empty} \rightarrow 8$

Final table

Index	Value
0	9679
1	4321
2	—
3	1323
4	6173
5	4344
6	—
7	—
8	1989
9	4199

D. Double hashing

Second

$h_2(x) = j - (x \text{ modulo } p)$

probe formula

$h(x) = (h(x) + i \cdot h_2(x)) \text{ modulo } 10$

Insertions

4321:

$h = 1 \rightarrow \text{index } 1$

1323:

$h = 1 \rightarrow \text{index } 3$

6173 $h = 3 \rightarrow \text{collision}$

~~4344~~

4199 $h = 9 \rightarrow \text{index } 9$

4344 $h = 4 \rightarrow \text{index } 4$

9679: $h = 9 \rightarrow \text{collision}$

9679 $h = 9 \rightarrow 5 \text{ index } 5$

Final double hashing table

Index	Value
0	—
1	4321
2	—
3	1323
4	4344
5	9679
6	—
7	—
8	6173
9	4199